eduX: Development Plan and Build Checklist

Oct 26, 2026 · @tafadzwa usayi

1. How to use this checklist

eduX is built module by module in the order of section 2, and a module is finished only when every item in it is ticked and its exit test passes. This document extends the DocHumanizer plan (now called eduX) and adds a Turnitin-style similarity engine (M8) and report (M9).

Item format. Every item reads: `ID` what to build, then the endpoint or component, then **Test:** how you prove it works. Tick the box only after the test passes.

ID scheme. `M4.7` = module 4, item 7. Tests are named after the item ID (for example `test_M4_7`) so each tick maps to one automated test.

Definition of done for an item

1. Code merged with a review.
2. The named test exists in CI and passes.
3. The endpoint appears in the OpenAPI file with request and response examples.
4. Errors are handled: invalid input returns a clear 4xx, failures are logged with a trace ID.
5. Plan limits and permissions are enforced where relevant.

Test types used below

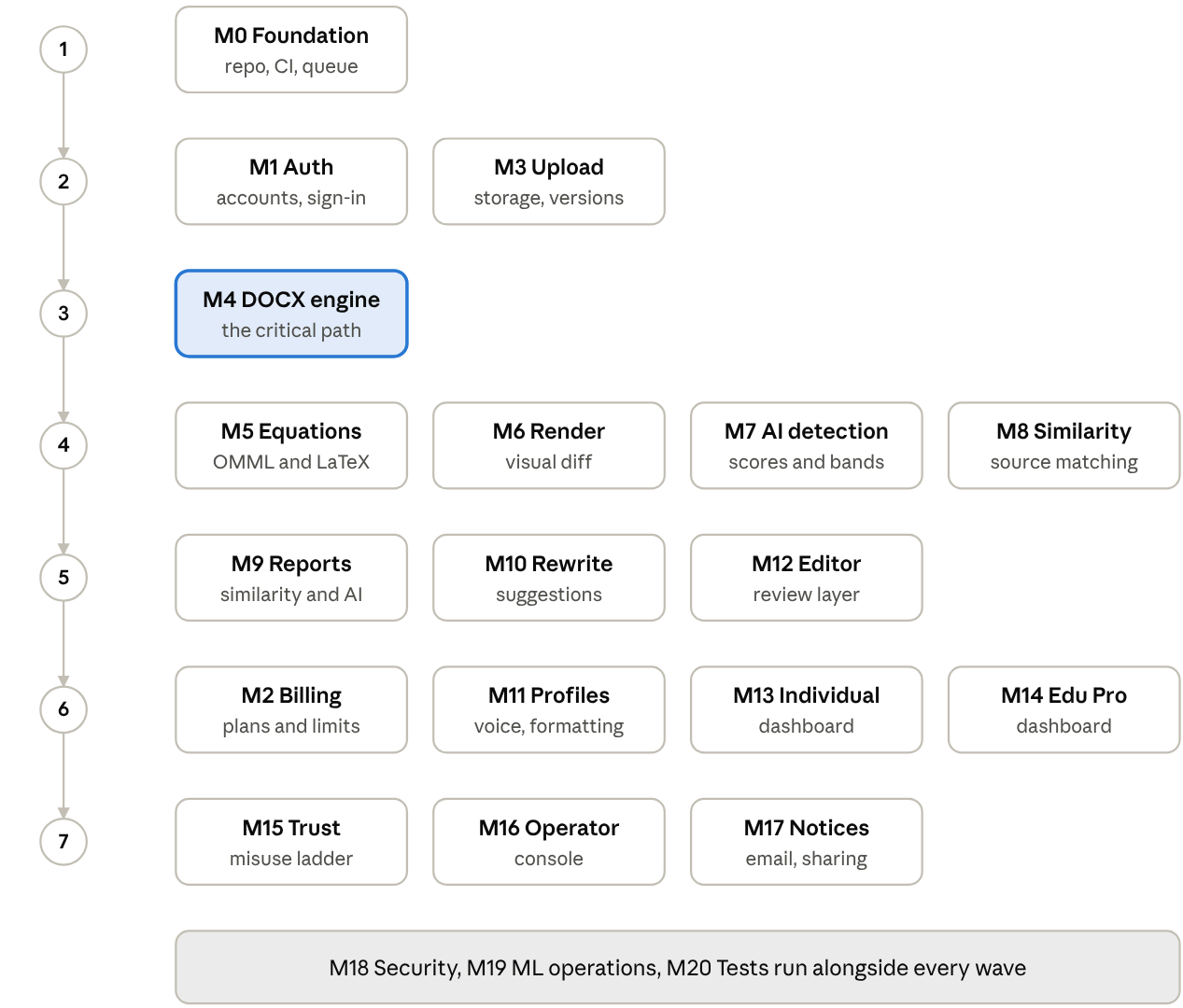
Tag	Meaning
Unit	One function or class, no network
API	HTTP call against a running service, checks status code and body
E2E	Browser test of a full user flow (Playwright)
Fidelity	Uses the DOCX corpus; untouched XML must be identical
Eval	Model metric on the frozen evaluation set must meet its gate

Tag	Meaning
Load	Meets a latency or throughput target under load
Security	Scan, penetration check or permission test

Conventions for endpoints. All paths start with `/v1`, use JSON, require a bearer token except where marked public, and return `{ error: { code, message } }` on failure. Long jobs return `202` with a job ID that you poll at `GET /v1/jobs/:id` or follow over server-sent events.

2. Build order

Build in seven waves, top to bottom; modules in the same wave can be developed in parallel by different people. Wave 3, the DOCX engine, is the critical path: detection, similarity, rewriting and the editor all read and write the document through it.



build order · 7 waves, 18 modules

Tick the checklist module by module in this order. Do not start a wave's exit test until the previous wave's exit tests pass, and keep the security, ML operations and test modules moving the whole time (they appear in every wave's pull requests).

M0. Foundation and infrastructure

Goal: a repository where any service can be built, tested and deployed in one command.
Exit test: a fresh clone runs `make dev` and the sample health checks pass.

- [ ] **M0.1** Monorepo with the layout from the plan (apps, services, ml, packages, infra, tests, docs). Test: CI builds every package from a clean checkout.

- [ ] **M0.2** Docker Compose with Postgres, Redis and MinIO. Test: `docker compose up` brings all three to healthy.
- [ ] **M0.3** Database migrations tool and baseline schema. Test: migrate up and down on an empty database without errors.
- [ ] **M0.4** OpenAPI contract file and generated typed clients. Test: CI fails if the contract and code disagree.
- [ ] **M0.5** `GET /v1/health` and `GET /v1/ready` on every service. Test (API): returns 200 with dependency status; returns 503 when Postgres is stopped.
- [ ] **M0.6** Structured logging with a trace ID on every request. Test: one upload produces the same trace ID in the API, queue and worker logs.
- [ ] **M0.7** Job queue with priorities, retries and a dead-letter queue. Test: a failing job retries 3 times, then lands in the dead-letter queue.
- [ ] **M0.8** `GET /v1/jobs/:id` and a server-sent events stream for job progress. Test (API): a job moves queued, running, done and the stream emits each state.
- [ ] **M0.9** Config and secrets management (env files locally, a secrets manager in cloud). Test: the app refuses to start when a required secret is missing.
- [ ] **M0.10** CI pipeline: lint, type check, unit tests, API tests, build. Test: a deliberately broken commit is blocked.
- [ ] **M0.11** Staging and production environments with Terraform. Test: `terraform plan` on staging shows no drift after apply.
- [ ] **M0.12** Feature flag system. Test: flipping a flag changes behaviour without a deploy.
- [ ] **M0.13** Error tracking and metrics dashboards. Test: a thrown test error appears in the tracker within one minute.
- [ ] **M0.14** Architecture decision record template and the first five ADRs. Test: the docs folder contains them and CI checks the front matter.

M1. Auth and accounts

Goal: secure sign-up, sign-in and account types (Individual, Edu Professional). **Exit test:** a new user signs up, verifies email, chooses an account type and reaches an empty dashboard.

- [ ] **M1.1** Email and password sign-up with strong hashing (Argon2). `POST /v1/auth/register`. Test (API): duplicate email returns 409; password under policy returns 422; hash is never returned.

- [ ] **M1.2** Email verification. `POST /v1/auth/verify`, `POST /v1/auth/resend`. Test (API): unverified users cannot upload; an expired token returns 410.
- [ ] **M1.3** Sign-in with access and refresh tokens. `POST /v1/auth/login`, `POST /v1/auth/refresh`, `POST /v1/auth/logout`. Test (API): refresh rotates the token and the old one stops working.
- [ ] **M1.4** Google sign-in (OIDC). `GET /v1/auth/google`, `GET /v1/auth/google/callback`. Test (E2E): a new Google user lands on plan choice.
- [ ] **M1.5** Password reset. `POST /v1/auth/forgot`, `POST /v1/auth/reset`. Test (API): a reset token works once only.
- [ ] **M1.6** Two-factor authentication (authenticator app). `POST /v1/auth/2fa/enable`, `POST /v1/auth/2fa/verify`. Test (API): login without the code is refused after enabling.
- [ ] **M1.7** Account type selection at sign-up: Individual or Edu Professional. `PATCH /v1/me/account-type`. Test (API): the type is stored and changes the limits returned by `GET /v1/usage`.
- [ ] **M1.8** Profile. `GET /v1/me`, `PATCH /v1/me` (name, time zone, country, institution). Test (API): time zone change moves the daily reset time.
- [ ] **M1.9** Sessions and devices list. `GET /v1/me/sessions`, `DELETE /v1/me/sessions/:id`. Test (API): a revoked session gets 401 on its next request.
- [ ] **M1.10** Rate limiting and lockout on login. Test (Security): 10 wrong passwords in a minute locks the account for a period and notifies the user.
- [ ] **M1.11** Role model: user, operator, admin, with permission checks as middleware. Test (API): a normal user gets 403 on every `/v1/ops/*` endpoint.
- [ ] **M1.12** Account deletion and data export. `DELETE /v1/me`, `POST /v1/me/export`. Test (API): after deletion the user's rows and files are queued for removal and sign-in fails.
- [ ] **M1.13** Frontend: sign-up, sign-in, verify, reset and plan-choice screens with accessibility checks. Test (E2E): each flow completes with the keyboard only.

M2. Plans, billing and usage limits

Goal: the Individual ($5 per month), Edu Pro ($15 per month) and pay-per-paper ($5) plans work end to end, and limits are enforced at the gateway. **Exit test:** a test card buys each plan, limits change accordingly, and a failed payment downgrades after the grace period.

- [ ] **M2.1** Plan catalogue stored in the database (price, caps, features) and editable by operators. `GET /v1/plans`. Test (API): changing a cap changes enforcement with no deploy.
- [ ] **M2.2** Checkout for subscriptions. `POST /v1/billing/checkout` returns a hosted payment link. Test (E2E): a test card completes checkout and the plan becomes active.
- [ ] **M2.3** Payment processor webhook with signature verification and idempotency. `POST /v1/billing/webhook`. Test (API): replaying the same event twice changes nothing; a bad signature returns 400.
- [ ] **M2.4** Pay-per-paper purchase that creates a 30-day licence on one document. `POST /v1/billing/paper-licence`. Test (API): the licence expires on day 30 and blocks new edits.
- [ ] **M2.5** Upgrade, downgrade and cancel with proration. `POST /v1/billing/change-plan`, `POST /v1/billing/cancel`. Test (API): upgrade takes effect immediately; downgrade takes effect at renewal.
- [ ] **M2.6** Failed payment handling with a 3-day grace period and reminder emails. Test: the account moves to read-only after the grace period and back to active after payment.
- [ ] **M2.7** Invoices and receipts as PDF. `GET /v1/billing/invoices`, `GET /v1/billing/invoices/:id/pdf`. Test (API): the PDF shows the right amount, tax line and plan.
- [ ] **M2.8** Usage counters per user per day (words scored, suggestions, uploads) kept in Redis and persisted to Postgres. `GET /v1/usage`. Test (Unit): counters reset at midnight in the user's time zone.
- [ ] **M2.9** Limit enforcement middleware returning `429` with the limit name and reset time. Test (API): the next request after the cap fails with a clear message, and a different user is unaffected.
- [ ] **M2.10** Active-papers cap and archive-to-free-a-slot. `POST /v1/documents/:id/archive`. Test (API): archiving the oldest paper allows a new upload.
- [ ] **M2.11** Free trial of 3 days or one free scored page. Test (API): the trial cannot be restarted on the same email, card or device fingerprint.
- [ ] **M2.12** Annual billing with discount and a local-payment option for supported markets. Test (E2E): a mobile-money or local-card sandbox payment activates the plan.
- [ ] **M2.13** Frontend: pricing page, checkout, billing settings, usage meter component. Test (E2E): the meter updates after each suggestion and shows the upgrade prompt at 90 percent of a cap.

- [ ] **M2.14** Tax and currency handling for the launch market. Test: a sample invoice in each launch currency matches the expected total.

M3. Upload and storage

Goal: safe upload of .docx files, versioned storage and document lifecycle. **Exit test:** a 40 MB document uploads, scans clean, is stored encrypted with version 1, and can be downloaded byte-identical.

- [ ] **M3.1** Direct-to-storage upload with a signed URL. `POST /v1/documents/upload-url`. Test (API): the URL expires after a few minutes and cannot be used for another path.
- [ ] **M3.2** Upload confirmation that creates the document and a scan job. `POST /v1/documents`. Test (API): returns 202 with a job ID; the document shows status `scanning`.
- [ ] **M3.3** File validation by content, not extension: valid zip, expected OOXML parts, size cap of 50 MB, .docx and .dotx only. Test (Security): a renamed .exe, a zip bomb and a password-protected file are all rejected with clear messages.
- [ ] **M3.4** Antivirus scan and macro stripping on the working copy. Test (Security): the EICAR test file is blocked; a macro-enabled file is cleaned.
- [ ] **M3.5** Encrypted storage with per-tenant keys and an immutable original. Test: the original's checksum never changes after edits.
- [ ] **M3.6** Document metadata. `GET /v1/documents`, `GET /v1/documents/:id`, `PATCH /v1/documents/:id` (title, status, due date, folder). Test (API): a user cannot read another user's document (404).
- [ ] **M3.7** Own-paper declaration on upload for Individual accounts. Field `ownerDeclaredOwn` in `POST /v1/documents`. Test (API): the value is stored and is available to the trust service.
- [ ] **M3.8** Version history. `GET /v1/documents/:id/versions`, `POST /v1/documents/:id/versions/:vid/restore`. Test (API): restoring creates a new version rather than deleting history.
- [ ] **M3.9** Download. `GET /v1/documents/:id/download?version=` with an audit log entry. Test: downloaded file opens in Word with no repair prompt.
- [ ] **M3.10** Archive, delete and restore from trash for 30 days. `POST /v1/documents/:id/archive`, `DELETE /v1/documents/:id`, `POST /v1/documents/:id/restore`. Test (API): permanent removal deletes files, rows, embeddings and cached renders.

- [ ] **M3.11** Duplicate-upload detection by file hash. Test: uploading the same file twice offers to open the existing document.
- [ ] **M3.12** Frontend: upload zone with progress, cancel, error states and the structure-scan summary. Test (E2E): a failed scan shows a plain-language reason and a retry button.

M4. DOCX engine

Goal: open any real Word document, map every sentence to its XML, change only what the user accepts, and export without damage. This is the highest-risk module, so its fidelity tests gate every release. **Exit test (Fidelity):** on the whole corpus, a no-op open and export leaves every XML part byte-identical, and a single-sentence edit changes only that paragraph.

- [ ] **M4.1** Fidelity corpus: start with 100 real documents (theses, reports, forms, tables, footnotes, comments, tracked changes, right-to-left text), grow to 500. Test: the runner reports pass or fail per document.
- [ ] **M4.2** Package reader for document.xml, styles, numbering, footnotes, endnotes, headers, footers, comments, media and relationships. Test (Unit): all parts load and re-save identically.
- [ ] **M4.3** Structure map with stable IDs for paragraphs, tables, cells, list items, runs, fields, images and section breaks. `GET /v1/documents/:id/structure`. Test (API): IDs are the same after re-parsing the same file.
- [ ] **M4.4** Run model that merges and splits runs without changing appearance. Test (Unit): applying the model and re-rendering gives the same pixels.
- [ ] **M4.5** Sentence segmentation that handles abbreviations, citations like (Smith, 2021), decimals, bullet lists and equations. Test (Unit): a set of 300 hard sentences splits correctly at 98 percent or better.
- [ ] **M4.6** Sentence-to-run mapping with character offsets. Test (Unit): every sentence's text equals the concatenated text of its runs.
- [ ] **M4.7** Word-level diff patcher that keeps unchanged words in their original runs and formatting. `POST /v1/sentences/:id/apply`. Test (Fidelity): editing one sentence leaves all other paragraphs byte-identical.
- [ ] **M4.8** Anchor safety: refuse or work around edits that would destroy hyperlinks, bookmarks, comment ranges or others' tracked changes. Test (Unit): each anchor type survives an edit nearby and blocks an edit across it.

- [ ] **M4.9** Protected atoms: tokenise equations, citations, tables, images, fields, cross-references, quoted blocks and numbers, restore them exactly. Test (Unit): tokenise then restore returns identical XML.
- [ ] **M4.10** Table support: edit text inside cells, lock structure, handle merged cells and nested tables. Test (Fidelity): table borders, widths and merges are unchanged after cell edits.
- [ ] **M4.11** Lists and numbering preserved, including multi-level and restarted lists. Test (Fidelity): list numbers are unchanged after edits.
- [ ] **M4.12** Footnotes, endnotes, headers and footers are readable and editable. Test (Fidelity): footnote numbering is unchanged.
- [ ] **M4.13** Event log: every accept, reject, undo and manual edit is an event; document state can be rebuilt from events. `POST /v1/documents/:id/undo`, `POST /v1/documents/:id/redo`. Test (API): ten edits then ten undos return the original bytes.
- [ ] **M4.14** Manual edit endpoint for typing directly into a sentence. `PATCH /v1/sentences/:id`. Test (API): text is patched into runs and a new version is recorded.
- [ ] **M4.15** Lock and ignore flags per sentence. `POST /v1/sentences/:id/lock`, `POST /v1/sentences/:id/ignore`. Test (API): locked sentences are skipped by scoring and rewriting.
- [ ] **M4.16** Tracked-change export: each accepted edit written as an insertion and deletion by the user. `POST /v1/documents/:id/export?mode=tracked`. Test: Word shows the changes and Accept All gives the same text as clean export.
- [ ] **M4.17** Clean export. `POST /v1/documents/:id/export?mode=clean`. Test (Fidelity): output opens in Word and LibreOffice with no repair prompt.
- [ ] **M4.18** Fidelity report listing changed paragraphs and confirming untouched ones. Included in the export response. Test (API): the changed-paragraph count equals the number of accepted edits.
- [ ] **M4.19** Performance: a 100-page document is parsed in under 10 seconds. Test (Load): measured on the corpus's largest files.
- [ ] **M4.20** Safe XML handling: no external entities, size limits, timeouts. Test (Security): a malicious XML file cannot read files or hang the worker.

M5. Equations (OMML and LaTeX)

Goal: equations display and edit as LaTeX and return to Word as native equations. **Exit test:** 95 percent of the equations in the test set round-trip with identical rendered output, and the rest stay locked and unchanged.

- [ ] **M5.1** Equation test set of 300 formulas (fractions, roots, sums, integrals, matrices, aligned multi-line, chemistry, custom operators). Test: the set is stored with expected LaTeX and rendered images.
- [ ] **M5.2** OMML extraction with stable atom IDs. `GET /v1/documents/:id/equations`. Test (API): the list matches the equation count from the structure scan.
- [ ] **M5.3** OMML to MathML to LaTeX conversion. Test (Unit): the converted LaTeX matches the expected string on at least 95 percent of the set.
- [ ] **M5.4** LaTeX to MathML to OMML conversion. `PUT /v1/equations/:id` with `{ latex }`. Test (API): the saved equation opens in Word as an editable equation.
- [ ] **M5.5** Round-trip safety check: convert, convert back, render both, compare. Test: any mismatch sets the equation to `locked` and keeps the original OMML.
- [ ] **M5.6** Inline versus display equations kept as they were. Test (Fidelity): an inline equation stays inline after an edit.
- [ ] **M5.7** Equation numbering and cross-references untouched. Test (Fidelity): equation numbers and references are identical after edits to nearby text.
- [ ] **M5.8** LaTeX validation with a clear error message. Test (API): invalid LaTeX returns 422 with the position of the error.
- [ ] **M5.9** Frontend: formulas rendered with KaTeX, click to open a LaTeX box with live preview and an Apply button. Test (E2E): editing a fraction updates the page preview and the exported file.
- [ ] **M5.10** Equations excluded from detection, similarity and rewriting, and shown as placeholders in reports. Test (Unit): the text sent to the models contains tokens, never formula text.

M6. Render and visual diff

Goal: show the document exactly as it looks, and prove that edits changed nothing else. **Exit test:** the visual diff of an untouched page against its original is under the pixel threshold on the whole corpus.

- [ ] **M6.1** Sandboxed LibreOffice render workers with time and memory limits. Test (Security): a malformed file cannot exceed limits or reach the network.
- [ ] **M6.2** Page images and thumbnails per version. `GET /v1/documents/:id/pages/:n?version=`. Test (API): the image is returned in under 2 seconds for a cached page.
- [ ] **M6.3** Render cache keyed by version and page; only changed pages re-render. Test: editing one paragraph re-renders only its page and the pages that reflow after it.
- [ ] **M6.4** Font handling: common Word fonts installed or substituted with metric-compatible fonts, and a warning when a font is missing. Test (Fidelity): the page count matches Word's for 95 percent of the corpus.
- [ ] **M6.5** Visual diff between original and edited versions with a threshold. `POST /v1/documents/:id/compare`. Test (API): untouched pages report zero difference; an edited page reports its changed regions.
- [ ] **M6.6** Layout damage detection: table overflow, orphan headings, missing images, page count change. Test: seeded faults in test documents are each detected.
- [ ] **M6.7** Vision-model structure helper for scanned or fake-structure documents (spaces for indents, text boxes for columns), suggested structure confirmed by the user. `POST /v1/documents/:id/structure/suggest`. Test (Eval): on 50 annotated hard documents, headings and tables are identified at 90 percent or better.
- [ ] **M6.8** Compare screen data: side-by-side images with a list of changes. Test (E2E): clicking a change scrolls both panes to it.

M7. AI detection

Goal: calibrated scores at document, paragraph and sentence level, streamed to the editor. **Exit test (Eval):** meets the release gates on the frozen set: false-positive rate under 1 percent on human text at the highest-confidence band, reported separately for native, non-native, formal-academic and technical writers.

- [ ] **M7.1** Data pipeline: human corpus (licensed, open, consented) and machine corpus from several open models at several temperatures and styles; hard negatives (grammar-edited human text, paraphrased machine text). Test: a data card exists for every set and train, validation and frozen test sets never overlap.
- [ ] **M7.2** Baseline classifier (fine-tuned encoder). Test (Eval): AUROC recorded on the frozen set and stored in the model registry.
- [ ] **M7.3** Reference-model features: per-token perplexity, burstiness and rank features. Test (Unit): features are deterministic for the same text and model version.

- [ ] **M7.4** Stylometric features (sentence length variance, punctuation rhythm, vocabulary richness, transition density). Test (Unit): feature vector shape and ranges are validated.
- [ ] **M7.5** Ensemble with a context smoother over neighbouring sentences. Test (Eval): the ensemble beats the best single model on AUROC and false-positive rate.
- [ ] **M7.6** Calibration by writer group (isotonic or Platt). Test (Eval): calibration error under the agreed limit for each group.
- [ ] **M7.7** Bands (likely human, uncertain, likely machine) with a configurable threshold. Test (Unit): band edges match the config; the uncertain band is never empty on mixed text.
- [ ] **M7.8** Skip rules: headings, lists, quotes, short sentences (under about 8 words), equations, references, mostly-numeric text. Test (Unit): skipped nodes return `not_scored` with a reason.
- [ ] **M7.9** Reason chips (uniform rhythm, generic phrasing, stock transitions, low specificity). Test (Eval): chips agree with human annotators on a sample of 200 sentences at the agreed rate.
- [ ] **M7.10** Scoring endpoints. `POST /v1/documents/:id/score` (202 with job), `GET /v1/documents/:id/scores` (streamed by paragraph), `GET /v1/sentences/:id/score`. Test (API): the first paragraph score arrives within 5 seconds on a 30-page document.
- [ ] **M7.11** Cache and re-score only changed paragraphs. Test: editing one sentence triggers scoring of one paragraph only.
- [ ] **M7.12** Robustness set: paraphrased, translated and lightly edited machine text; adversarial rounds feed back into training. Test (Eval): detection rate on the set is recorded for each model version.
- [ ] **M7.13** Document summary with plain-language explanation and a statement of uncertainty. `GET /v1/documents/:id/score-summary`. Test (API): a mostly-formula document returns `too_little_prose` rather than a score.
- [ ] **M7.14** Detector limits message everywhere a score is shown: scores are estimates, not proof. Test (E2E): the notice is visible on the editor, the report and the export.
- [ ] **M7.15** Serving: batching, quantisation, autoscaling. Test (Load): 50 concurrent documents stay within latency targets.

M8. Similarity engine (Turnitin-style matching)

Goal: find passages in a document that match other sources, report an overall similarity percentage, and link each match to its source. **Exit test (Eval):** on a synthetic plagiarism test set (copied, lightly edited and paraphrased passages), recall and precision meet the agreed targets, and a document compared with itself reports near 100 percent.

Honest scope. A tool's matching power depends on the size of its source database. Turnitin's strength is a large proprietary store of student papers and licensed content, which eduX cannot copy on day one. The plan starts with three source pools and states which pools were searched in every report, so users are never told "no matches" about sources the system never checked.

Source pool	Content	Notes
Open scholarly	arXiv, PubMed Central open access, CORE, OpenAlex abstracts and open full text, Wikipedia, public-domain books	Free to index; large coverage of academic text
Web index	Pages from an open web crawl, filtered for quality and text content	Needs crawl and storage budget; start with a sampled index
eduX repository	Documents users or institutions opt in to add	Consent only, per-institution private pools, never public
Licensed (later)	Publisher and database partners through agreements	Phase V2

- [ ] **M8.1** Source ingestion pipeline: fetch, extract text, clean, deduplicate, store with licence and origin. Test: each ingested source records URL, date fetched, licence and a content hash.
- [ ] **M8.2** Text normalisation (case, punctuation, hyphenation, numbers, stop-word handling) shared by sources and submissions. Test (Unit): the same passage normalises identically from .docx, PDF text and HTML.
- [ ] **M8.3** Fingerprinting: word n-gram shingles (for example 6 to 8 words) with winnowing to select stable fingerprints. Test (Unit): copying a 50-word passage with one word changed still shares fingerprints.

- [ ] **M8.4** Inverted index of fingerprints with source and position, sharded and compressed. Test (Load): lookup of a 10,000-word document completes within the target time on the test index.
- [ ] **M8.5** Near-duplicate detection with MinHash and locality-sensitive hashing for whole-document and large-passage matches. Test: a 90 percent-identical copy is found in the top result.
- [ ] **M8.6** Match extension and alignment: grow seed matches into maximal spans, merge nearby spans, drop very short matches (configurable, default about 8 words). Test (Unit): a copied paragraph with two small edits is one match, not three.
- [ ] **M8.7** Paraphrase layer: sentence embeddings with a vector index, then a cross-encoder to confirm meaning match. Test (Eval): recall on the paraphrased part of the test set meets target with a low false-match rate.
- [ ] **M8.8** Exclusions: quoted text with citations, bibliography and references, template text (cover page, instructions), common phrases, small matches. `POST /v1/documents/:id/similarity/exclusions`. Test (API): excluding the bibliography lowers the percentage and the change is listed in the report.
- [ ] **M8.9** Citation awareness: matched passages inside quotation marks with a nearby citation are labelled `quoted and cited` rather than `uncited match`. Test (Unit): the label depends on both the quotes and the citation.
- [ ] **M8.10** Similarity percentage definition and calculation (matched words not excluded divided by total counted words), with per-source percentages that avoid double counting when sources overlap. Test (Unit): sources that overlap on the same words count those words once in the total.
- [ ] **M8.11** Source ranking and grouping (primary source, other sources, repository, web, scholarly). Test: the report lists the top sources first with their match counts.
- [ ] **M8.12** Self-similarity and reuse checks: the same text repeated inside a document, and reuse of the user's own earlier submissions in the repository. Test (API): a repeated paragraph is flagged as internal repetition.
- [ ] **M8.13** Class and folder comparison for Edu Pro (collusion check across documents in a folder). `POST /v1/folders/:id/similarity/compare`. Test (API): two papers sharing a copied section appear as a pair with the matched passages.
- [ ] **M8.14** Similarity endpoints. `POST /v1/documents/:id/similarity` (202 with job), `GET /v1/documents/:id/similarity` (summary), `GET /v1/documents/:id/similarity/matches` (passages with offsets and source IDs), `GET /v1/documents/:id/similarity/sources`. Test (API): offsets line up with the sentence IDs from M4.

- [ ] **M8.15** Source viewer: show the matched snippet and a link to the original where licence allows; never reproduce a full copyrighted source. `GET /v1/similarity/sources/:id/snippet?matchId=`. Test (API): the snippet is capped at a limited length around the match.
- [ ] **M8.16** Repository opt-in controls: per-document and per-account consent, institution private pool, removal request. `PATCH /v1/documents/:id/repository`, `DELETE /v1/repository/:docId`. Test (API): a removed document no longer appears as a match source within the next index refresh.
- [ ] **M8.17** Index refresh and re-crawl jobs with admin controls. `POST /v1/ops/sources/refresh`. Test: new sources appear in match results after the refresh.
- [ ] **M8.18** Coverage statement included with every result (which pools, index date, size). Test (E2E): the report shows the coverage panel.
- [ ] **M8.19** Evaluation set built from PAN-style plagiarism benchmarks and our own seeded documents (verbatim, edited, paraphrased, translated, shuffled). Test (Eval): precision and recall are stored per model and index version, and a drop blocks release.
- [ ] **M8.20** Cost and latency control: cache by document hash, incremental re-check of changed paragraphs only. Test: re-checking after one sentence edit takes a few seconds, not a full pass.

M9. Reports (similarity, AI and combined)

Goal: an eduX report like Turnitin's: an overall similarity percentage, colour-highlighted matched text, a source list, and an AI-writing section, shareable and exportable as PDF.
Exit test: a 30-page document produces a report in under 90 seconds, the PDF matches the on-screen report, and every highlight links to its match and source.

Report contents

Section	What it shows
Header	Document title, owner or client, date, plan, report ID, pools searched
Similarity summary	Overall percentage, number of sources, longest match, breakdown by pool, exclusions used

Section	What it shows
Highlighted text	The document with matches coloured by source and a number badge for each match
Source list	Ranked sources with match percentage, type (web, scholarly, repository), link or reference
Match detail	Selected match side by side with the source snippet
AI writing section	Document score band, paragraph heat map and the limits notice (scores are estimates)
Integrity notes	Quoted and cited passages, internal repetition, formatting oddities
Methods and coverage	How the percentage is calculated, index date and size, model versions

- [ ] **M9.1** Report data model: report, report version, settings snapshot (exclusions, thresholds), model and index versions. Test (Unit): regenerating a report with the same settings and versions gives the same numbers.
- [ ] **M9.2** Generate report. `POST /v1/documents/:id/reports` with `{ similarity: true, ai: true, exclusions }` (202 with job). Test (API): a report ID is returned and the job reaches done.
- [ ] **M9.3** Fetch report data. `GET /v1/reports/:id` (JSON), `GET /v1/reports/:id/matches`, `GET /v1/reports/:id/sources`. Test (API): the sum of per-source percentages is at or above the overall percentage only where sources overlap, as defined in M8.10.
- [ ] **M9.4** Interactive report viewer: highlighted document, source list, click a match to see the source side by side, filter by source type, toggle exclusions live. Test (E2E): toggling "exclude quotes" updates the percentage and highlights.
- [ ] **M9.5** Combined view: similarity and AI layers can be switched on and off over the same document. Test (E2E): the layers do not hide each other's marks.
- [ ] **M9.6** PDF export with a clean layout, page numbers, numbered match badges and the coverage and methods page. `GET /v1/reports/:id/pdf`. Test: the PDF has the same percentage and source list as the screen.

- [ ] **M9.7** Report sharing: read-only link, expiring, revocable, optional password, optional hide-sources mode for students. `POST /v1/reports/:id/share`, `DELETE /v1/reports/:id/share/:token`. Test (API): a revoked link returns 404 immediately.
- [ ] **M9.8** Before and after comparison: report on version 1 against version 2 with the change in percentage. `GET /v1/documents/:id/reports/compare?from=&to=`. Test (API): the delta equals the difference of the two reports.
- [ ] **M9.9** Rewrite from report: click a highlighted match and open the editor at that sentence with suggestions (rewrite, add quotation marks and citation placeholder). Test (E2E): applying a suggestion and re-checking lowers or keeps the match, never silently hides it.
- [ ] **M9.10** Report access and plan rules: Individual users see reports on their own papers; Edu Pro users see reports per client folder; report counts count against daily limits. Test (API): limits and permissions return the right codes.
- [ ] **M9.11** Report wording review: plain language, no accusation, a notice that similarity is not the same as plagiarism and that AI scores are estimates. Test (E2E): the notice appears on screen, in the PDF and on shared links.
- [ ] **M9.12** Report history and retention. `GET /v1/documents/:id/reports`. Test (API): reports are deleted with the document.
- [ ] **M9.13** Accessibility and print: colour-blind safe highlight palette with patterns, readable in greyscale. Test: automated contrast check and a manual greyscale print review.

M10. Rewrite and suggestions

Goal: three checked suggestions for any sentence, one-click auto-rearrange, and bulk review. **Exit test (Eval):** at least the agreed share of suggestions pass the meaning and constraint checks, and none introduce a new number, name or citation.

- [ ] **M10.1** Context builder: target sentence, paragraph, headings and a short document summary. Test (Unit): locked atoms appear only as tokens.
- [ ] **M10.2** Constraint extractor: numbers, dates, named entities, terminology list, citations. Test (Unit): the extractor finds every number and entity in a labelled set.
- [ ] **M10.3** Base rewriter serving an open instruction-tuned model with candidate strategies (restructure, vary rhythm, plain phrasing, split, merge). Test (API): returns at least three distinct candidates in under 3 seconds.

- [ ] **M10.4** Meaning check: embedding similarity plus two-way entailment. Test (Eval): candidates that change meaning are rejected on a labelled set at the agreed rate.
- [ ] **M10.5** Constraint check: every locked item is present and unchanged. Test (Unit): a candidate that drops a number is rejected.
- [ ] **M10.6** No-new-facts guard (new entities, numbers or claims not in the source). Test (Eval): seeded hallucination cases are caught.
- [ ] **M10.7** Re-scoring of each candidate in paragraph context and ranking. Test (API): candidates are returned with `meaningScore`, `scoreDelta` and a short note on what changed.
- [ ] **M10.8** Suggest endpoint. `POST /v1/sentences/:id/suggest` with `{ tone, formality, length, strength, keep[] }`. Test (API): changing the strength changes how much the sentence is rewritten.
- [ ] **M10.9** Apply endpoint. `POST /v1/sentences/:id/apply` with `{ suggestionId }` or `{ text }`. Test (API): the document version increments and the event is logged.
- [ ] **M10.10** Auto-rearrange for a sentence. `POST /v1/sentences/:id/auto`. Test (API): applies the top candidate and returns an undo token.
- [ ] **M10.11** Auto-rearrange for a paragraph with transition checks. `POST /v1/paragraphs/:id/auto`. Test (Eval): paragraph flow rating by human reviewers is not lower than the original on average.
- [ ] **M10.12** Bulk fix with a review queue. `POST /v1/documents/:id/bulk-suggest`, `GET /v1/documents/:id/review-queue`, `POST /v1/review-queue/:itemId/accept|skip`. Test (API): nothing is applied until an item is accepted.
- [ ] **M10.13** Feedback capture (accept, edit then accept, reject, thumbs, reason, "changed my meaning"). `POST /v1/suggestions/:id/feedback`. Test (API): feedback is stored with the model version and the opt-in flag.
- [ ] **M10.14** Rate limits and cost caps per plan on suggestion calls. Test (API): the cap returns 429 with the reset time.
- [ ] **M10.15** Suggestion cache for identical sentence and settings. Test: a repeated request returns the cached result.
- [ ] **M10.16** Report-aware suggestions: for a similarity match, offer quote-and-cite, paraphrase, or restructure. Test (Unit): a quoted and cited passage is never auto-paraphrased.

M11. Voice and formatting profiles

Goal: rewrites that sound like the user, and one-click formatting to a chosen style. **Exit test:** a voice profile built from two sample documents measurably moves rewrites closer to those samples than the default, and applying APA to a messy document yields only style-based formatting changes.

Voice profile

- [ ] **M11.1** Sample upload (2 to 3 documents) with consent text. `POST /v1/voice-profiles`. Test (API): fewer than the minimum word count returns a clear message.
- [ ] **M11.2** Style extraction: sentence length, vocabulary, punctuation habits, connective phrases, spelling variety (US or UK). Test (Unit): profile fields are filled for a sample set.
- [ ] **M11.3** Style vector and short style guide stored per profile. `GET /v1/voice-profiles/:id`. Test (API): the profile returns strength and sample count.
- [ ] **M11.4** Conditioning of the rewriter on the profile. Field `voiceProfileId` in `POST /v1/sentences/:id/suggest`. Test (Eval): held-out authorship matching is better with the profile than without.
- [ ] **M11.5** Voice match score per paragraph. Test (API): a paragraph written by the user scores higher than one from another author.
- [ ] **M11.6** Edu Pro: several profiles, each tied to a client with a consent flag. Test (API): a profile cannot be used on a folder it is not assigned to.
- [ ] **M11.7** Profile reset and deletion with vector removal. `DELETE /v1/voice-profiles/:id`. Test (API): vectors and samples are removed.

Formatting profiles

- [ ] **M11.8** Built-in profiles: APA 7, Harvard, MLA 9, plus a blank custom profile. `GET /v1/style-profiles`. Test (API): each profile lists its rules.
- [ ] **M11.9** Rules mapped to real Word styles (Heading 1 to 3, Body Text, Caption, Block Quote) and page setup. Test (Fidelity): output uses style definitions, not direct formatting.
- [ ] **M11.10** Formatting audit: list deviations from the chosen profile with locations. `POST /v1/documents/:id/format/audit`. Test (API): a seeded wrong font and spacing are reported.
- [ ] **M11.11** Preview and apply with before and after render. `POST /v1/documents/:id/format/preview`, `POST /v1/documents/:id/format/apply`. Test (API): only audited deviations are changed.

- [ ] **M11.12** Custom profile from an uploaded .dotx template. `POST /v1/style-profiles/import`. Test (API): the template's styles become rules.
- [ ] **M11.13** Folder-level assignment for Edu Pro and batch apply. `POST /v1/folders/:id/format/apply`. Test (API): a batch job reports success or failure per document.
- [ ] **M11.14** Reference list and citation-style checks (order, punctuation, missing year) without altering content. Test (Unit): findings are warnings and never auto-edits.

M12. Editor frontend

Goal: a Word-like editor with a review layer for AI scores, similarity matches, suggestions and formulas. **Exit test (E2E):** open a 30-page document, flag, fix and undo a sentence, edit an equation, run a similarity check, and export, all with the keyboard only.

- [ ] **M12.1** App shell, routing, authentication guard, error boundaries, design system tokens. Test (E2E): signed-out users are redirected to sign-in.
- [ ] **M12.2** Page-by-page document view built from the structure map and render images, virtualised for long documents. Test (Load): scrolling a 100-page document stays smooth.
- [ ] **M12.3** Sentence selection and hover with stable IDs. Test (E2E): clicking any sentence highlights exactly that sentence, including across formatting changes.
- [ ] **M12.4** Heat map layers: AI bands, similarity matches, both, or none; threshold slider; legend. Test (E2E): toggling a layer shows and hides only that layer.
- [ ] **M12.5** Right panel tabs: Suggestions, Score details, Similarity, Formatting, History. Test (E2E): selecting a matched sentence opens its match and source.
- [ ] **M12.6** Floating selection menu: Rewrite, Suggest, Auto, Edit by hand, Lock, Ignore. Test (E2E): each action calls the right endpoint.
- [ ] **M12.7** Suggestion preview on hover and Apply, Auto and Undo with a toast. Test (E2E): apply then undo restores the sentence and its formatting.
- [ ] **M12.8** Manual editing inside a sentence with run-safe patching. Test (E2E): typing in a bold phrase keeps it bold.
- [ ] **M12.9** Equation editing with KaTeX preview. Test (E2E): see M5.9.
- [ ] **M12.10** Streaming score updates over server-sent events with a progress indicator. Test (E2E): paragraph marks appear as scores arrive, with no layout jump.
- [ ] **M12.11** Left rail: outline, thumbnails and a heat strip. Test (E2E): clicking a heat strip segment scrolls to that location.

- [ ] **M12.12** Review queue UI for bulk fixes with keyboard shortcuts (J, K, A, S, U, L). Test (E2E): the shortcut sheet lists every shortcut and each works.
- [ ] **M12.13** Compare view and fidelity report display. Test (E2E): the changes list scrolls both panes.
- [ ] **M12.14** Export dialog: clean or tracked, report included or not. Test (E2E): the downloaded file matches the chosen mode.
- [ ] **M12.15** Autosave state, offline-error handling and conflict message when the document changed elsewhere. Test (E2E): a dropped connection shows a banner and recovers.
- [ ] **M12.16** Accessibility to WCAG 2.2 AA: keyboard navigation, screen-reader labels, focus order, colours with patterns. Test: automated checks pass and one manual screen-reader run.
- [ ] **M12.17** Performance targets: a 30-page document opens in under 4 seconds, first scores in under 5 seconds, apply updates the view in under 100 ms. Test (Load): measured in CI on a reference machine.

M13. Individual dashboard

Goal: one screen where an Individual user manages their papers, usage and plan. **Exit test (E2E):** a new Individual user uploads a paper, sees its status and scores, hits a soft limit, and upgrades.

- [ ] **M13.1** Dashboard summary endpoint. `GET /v1/dashboard`. Test (API): returns papers, usage, plan and recent activity in one call under 300 ms.
- [ ] **M13.2** Upload zone widget with drag and drop and progress. Test (E2E): see M3.12.
- [ ] **M13.3** My papers cards (title, status, AI band, similarity percentage, last edited) with open, rename, duplicate, archive, delete. Test (E2E): each action updates the card without a reload.
- [ ] **M13.4** Paper status (draft, in review, ready) and optional due date with a reminder. `PATCH /v1/documents/:id`. Test (API): a due date in the past shows overdue.
- [ ] **M13.5** Score and similarity trend per paper across versions. `GET /v1/documents/:id/trend`. Test (API): points exist for each scored version.
- [ ] **M13.6** Usage meter widget (words, suggestions, active papers, today and this month). Test (E2E): see M2.13.
- [ ] **M13.7** Voice profile card with add samples and strength. Test (E2E): adding a sample updates strength.

- [ ] **M13.8** Recent activity feed. `GET /v1/activity`. Test (API): the last 20 events appear in time order.
- [ ] **M13.9** Plan and billing card with upgrade to Edu Pro and single-paper purchase. Test (E2E): upgrade opens checkout and returns to the dashboard with the new plan.
- [ ] **M13.10** Own-paper confirmation prompt on upload, wording reviewed for courtesy. Test (E2E): the checkbox is required for Individual accounts.
- [ ] **M13.11** Personal tips card based on the user's own patterns (for example overused openers). `GET /v1/tips`. Test (API): each tip cites an example from the user's text.
- [ ] **M13.12** Empty, loading and error states for every widget, and mobile layout. Test (E2E): each state is reachable and readable at phone width.

M14. Edu Pro dashboard

Goal: manage many clients, students or classes, with batch tools, reports and returns.
Exit test (E2E): an Edu Pro user creates two folders, uploads five papers, runs a batch report, moves papers across the board, and returns an edited file by link.

- [ ] **M14.1** Folders and client labels. `POST /v1/folders`, `GET /v1/folders`, `PATCH /v1/folders/:id`, `DELETE /v1/folders/:id`. Test (API): deleting a folder asks what to do with its documents.
- [ ] **M14.2** Bulk move, label and archive of documents. `POST /v1/documents/bulk`. Test (API): a mixed result returns per-document success or error.
- [ ] **M14.3** Document board with columns Uploaded, In review, Returned, Closed and drag to change stage. `PATCH /v1/documents/:id/stage`. Test (E2E): dragging a card persists after reload.
- [ ] **M14.4** Queue and deadlines view with overdue alerts and client filter. `GET /v1/queue`. Test (API): sorted by due date with overdue first.
- [ ] **M14.5** Throughput and turnaround charts. `GET /v1/analytics/throughput?from=&to=`. Test (API): counts equal closed documents in the range.
- [ ] **M14.6** Quality snapshot (average AI-score drop per document, similarity change, suggestion acceptance rate) with per-client breakdown. `GET /v1/analytics/quality`. Test (API): numbers match the underlying events.
- [ ] **M14.7** Batch tools: score, similarity and format a whole folder with progress and a review queue. `POST /v1/folders/:id/batch`. Test (API): the cap on documents per batch is enforced by plan.
- [ ] **M14.8** Folder similarity comparison (collusion check). Test: see M8.13.

- [ ] **M14.9** Share and return: read-only review link or send the edited file to the owner's email. `POST /v1/documents/:id/return`. Test (API): the owner receives an email with an expiring link.
- [ ] **M14.10** Style profile library with assignment to folders. Test (E2E): documents uploaded into a folder pick up its profile.
- [ ] **M14.11** Voice profiles per client with consent flag. Test: see M11.6.
- [ ] **M14.12** Usage and limits panel with a limit-increase request. `POST /v1/limits/request`. Test (API): the request appears in the operator console.
- [ ] **M14.13** Billing view with receipts per month and a company name on invoices. Test (API): the invoice shows the saved company name.
- [ ] **M14.14** Priority queue for Edu Pro jobs. Test (Load): under load, Edu Pro jobs start before Individual jobs of the same age.
- [ ] **M14.15** CSV export of the document list and analytics. `GET /v1/analytics/export.csv`. Test: the CSV opens cleanly in a spreadsheet with correct columns.

M15. Trust and misuse

Goal: spot Individual accounts used for other people's papers and shared logins, and respond with a fair, graduated process. **Exit test:** replaying 30 days of seeded behaviour puts the right test accounts at the right ladder level, and every action shows the user a reason and an appeal path.

- [ ] **M15.1** Signal collectors: document metadata (author, last modified by, company), title-page entities, course codes, upload cadence, devices, locations, session overlap, payment fingerprints. Test (Unit): each collector writes a `risk_signals` row with type and value.
- [ ] **M15.2** Authorship style fingerprint per paper stored as a numeric vector, never as text. Test (Unit): the vector cannot be turned back into the document.
- [ ] **M15.3** Style-spread feature: distance between papers on one account over a rolling 30 days. Test (Unit): two documents by different authors give a larger spread than two by the same author.
- [ ] **M15.4** Rule engine for clear cases (for example four or more distinct authors in a month on Individual, or more than three simultaneous devices). Test (Unit): each rule fires on its seeded case and not on a clean case.

- [ ] **M15.5** Risk score model (gradient-boosted trees) with top reasons; starts rules-only until labelled cases exist. `GET /v1/ops/users/:id/risk`. Test (Eval): precision at each level is recorded on labelled cases.
- [ ] **M15.6** Enforcement ladder: level 0 none, 1 nudge, 2 confirm or upgrade, 3 throttle, 4 human review. Test (API): moving between levels changes the responses of upload endpoints as designed.
- [ ] **M15.7** User-facing messages with the reason and fix at each level. `GET /v1/me/notices`. Test (E2E): the wording is courteous and offers an Edu Pro upgrade or an appeal.
- [ ] **M15.8** Upgrade-to-Edu-Pro flow that resets the level. Test (API): after upgrade the user is at level 0 with Edu Pro caps.
- [ ] **M15.9** Appeals. `POST /v1/appeals`, `GET /v1/ops/appeals`, `POST /v1/ops/appeals/:id/decide`. Test (API): a decision restores or keeps the level, with a logged reason.
- [ ] **M15.10** Abuse case queue for reviewers with evidence panel (no document text unless the user opens a ticket). `GET /v1/ops/abuse-cases`. Test (Security): reviewers cannot open document text from this view.
- [ ] **M15.11** Free-trial farming, credential sharing, chargeback fraud, and scraping rules with device and payment fingerprints and bot checks. Test (Security): scripted sign-ups from one device are throttled.
- [ ] **M15.12** Fairness metrics: share of flags cleared on appeal (target under 5 percent), flags by country and plan. Test: the dashboard shows them and alerts above the limit.
- [ ] **M15.13** Disclosure: signals described in the terms and privacy notice, and shown in the account page. Test (E2E): the page lists what is collected and why.

M16. Operator console

Goal: an internal tool to run the business: users, billing, abuse, models, sources and limits. **Exit test (E2E):** an operator finds a user, changes their plan, refunds a payment, resolves an abuse case and rolls back a model, and every step appears in the audit log.

- [ ] **M16.1** Operator sign-in with mandatory two-factor and role checks. Test (Security): no operator route works without both.
- [ ] **M16.2** Overview (active users, sign-ups, revenue, conversion, churn, by plan). `GET /v1/ops/overview`. Test (API): totals match the database for a seeded month.

- [ ] **M16.3** Users: search, detail, change plan, suspend, impersonate with banner and audit entry. `GET /v1/ops/users`, `POST /v1/ops/users/:id/suspend`, `POST /v1/ops/users/:id/impersonate`. Test (API): every impersonation writes an audit row and expires in a short time.
- [ ] **M16.4** Billing tools: payments, failures, disputes, refunds, credits. `POST /v1/ops/payments/:id/refund`. Test (API): a refund updates the processor and the invoice.
- [ ] **M16.5** Abuse queue and appeals views. Test: see M15.9 and M15.10.
- [ ] **M16.6** Usage and cost: model calls, GPU hours, cost per user and per plan, alert when cost per user exceeds revenue. `GET /v1/ops/costs`. Test (API): a seeded heavy user triggers the alert.
- [ ] **M16.7** Model quality: false-positive rate by writer group, similarity precision and recall, suggestion acceptance, meaning-check failures, with version comparison and rollback. `POST /v1/ops/models/:id/rollback`. Test (API): rollback switches the serving version within one minute.
- [ ] **M16.8** Feedback review and training-set approval (opted-in data only). `POST /v1/ops/feedback/:id/approve`. Test (API): non-opted-in feedback cannot be approved.
- [ ] **M16.9** Source index management (pools, size, last refresh, takedown requests). `GET /v1/ops/sources`, `POST /v1/ops/sources/takedown`. Test (API): a takedown removes the source from matches after refresh.
- [ ] **M16.10** Fidelity monitor: export fidelity failures and visual-diff alerts with the failing document (access by ticket only). Test: a seeded failure shows up with its trace ID.
- [ ] **M16.11** Limits and plan editor with feature flags per cohort. `PATCH /v1/ops/plans/:id`, `POST /v1/ops/flags`. Test (API): a change applies within one minute.
- [ ] **M16.12** Content policy queue (reports, blocked uploads). Test (API): actions are logged and reversible.
- [ ] **M16.13** Audit log, filterable and exportable. `GET /v1/ops/audit`. Test (API): every `/v1/ops/*` write creates an entry with actor, target and time.
- [ ] **M16.14** Support tools: ticket link to a document with user-granted access for a limited time. `POST /v1/support/access-grants`. Test (Security): access ends on time and is visible to the user.

M17. Notifications and sharing

Goal: timely, controllable messages and safe sharing of documents and reports. **Exit test:** each event type produces an in-app and an email notification that respects the user's settings, and shared links obey their expiry and revocation.

- [ ] **M17.1** Notification service with templates and a delivery log. `GET /v1/notifications`, `POST /v1/notifications/:id/read`. Test (API): unread count changes after reading.
- [ ] **M17.2** Events: near a daily limit, limit reached, payment failed, plan changed, report ready, export ready, review link opened, shared document returned, abuse notice, appeal decided. Test: each event type is sent once, with the right template and no duplicates.
- [ ] **M17.3** Email delivery with a verified sending domain, bounce handling and unsubscribe for non-essential mail. Test: bounced addresses stop receiving mail and are flagged.
- [ ] **M17.4** Notification preferences. `GET /v1/me/notification-settings`, `PATCH /v1/me/notification-settings`. Test (API): disabling a channel stops it, except security and billing notices.
- [ ] **M17.5** Document share links with expiry, revoke, optional password and view-only mode. `POST /v1/documents/:id/share`. Test: see M9.7.
- [ ] **M17.6** Share link access log (opens, time, country). `GET /v1/shares/:id/log`. Test (API): each open adds an entry without storing personal data beyond the policy.
- [ ] **M17.7** Return to owner with an email and a download link. Test: see M14.9.
- [ ] **M17.8** Rate limits on notifications to prevent spam. Test: bulk actions send one summary email rather than many.

M18. Security and compliance

Goal: protect unpublished papers and meet privacy law. **Exit test (Security):** an external penetration test finds no critical or high issues open, and a data deletion request removes all of a user's data within the stated window.

- [ ] **M18.1** Encryption at rest (AES-256, per-tenant keys) and in transit (TLS 1.3, HSTS). Test (Security): a scan shows no weak protocols and all buckets are private.
- [ ] **M18.2** Sandboxed workers with no outbound network, CPU, memory and time limits. Test (Security): a hostile file cannot reach the network or exhaust the node.

- [ ] **M18.3** Authorisation tests on every endpoint (own data only, role checks, folder permissions). Test (Security): an automated matrix proves a user cannot read, change or delete another user's resources.
- [ ] **M18.4** Input validation, output encoding, CSRF protection, secure headers and a content security policy. Test (Security): a standard web scanner reports no high findings.
- [ ] **M18.5** Dependency and container scanning in CI. Test: a known-vulnerable package fails the build.
- [ ] **M18.6** Secrets rotation and key management. Test: rotating a key does not break reads of old documents.
- [ ] **M18.7** Audit logging for downloads, impersonation, exports, share creation and operator actions. Test: each listed action creates an entry.
- [ ] **M18.8** Data deletion pipeline covering files, rows, embeddings, caches, search indexes and the similarity repository, with backups ageing out on a stated schedule. `DELETE /v1/me`. Test: a deleted test account leaves nothing behind in any store.
- [ ] **M18.9** Training-data consent: opt-in flag, identifier removal, and a list of included records for deletion at retraining. Test (Unit): non-opted-in text never reaches the training export.
- [ ] **M18.10** No document text sent to outside AI providers; models run on our own infrastructure. Test (Security): outbound traffic from model workers is blocked and logged.
- [ ] **M18.11** Privacy policy, terms, responsible-use terms, cookie banner, data processing agreement and a subprocessors list. Test: the sign-up flow records acceptance with version and time.
- [ ] **M18.12** Regional compliance review: GDPR and UK GDPR, Zimbabwe's data protection law, other launch-market laws, and CCPA. Test: a checklist signed off by a legal reviewer.
- [ ] **M18.13** Copyright and takedown process for the similarity source index and for repository content. Test: a takedown request is handled within the stated time in a drill.
- [ ] **M18.14** Backups with restore drills. Test: a restore into a clean environment succeeds and passes smoke tests.
- [ ] **M18.15** Incident response plan and an on-call rota. Test: a tabletop exercise is completed and its actions are tracked.

M19. ML operations and evaluation

Goal: train, evaluate, ship and roll back every model safely and repeatably. **Exit test:** a new model version goes from training to shadow to canary to full release through the pipeline with its metrics recorded, and a worse version is blocked automatically.

- [ ] **M19.1** Experiment tracking and model registry (MLflow or Weights and Biases) with data version, code version and metrics for every model. Test: any served model can be traced to its data and code.
- [ ] **M19.2** Data versioning (DVC or lakeFS) and data cards for the AI-detection, similarity, rewrite, voice and trust data sets. Test: reproducing a training run from a registry entry gives metrics within tolerance.
- [ ] **M19.3** Frozen evaluation sets, one per model, stored read-only and never used for training. Test: a CI check fails if evaluation IDs appear in training data.
- [ ] **M19.4** Release gates in code: detector false-positive rate by group, AUROC and calibration; similarity precision and recall; rewrite meaning and constraint pass rates; trust precision per level. Test: a model that misses any gate cannot be promoted.
- [ ] **M19.5** Fine-tuning scripts (LoRA or QLoRA) for the rewriter and encoder training scripts for the detector. Test: one command reproduces the latest model from its config.
- [ ] **M19.6** Preference tuning from accept, edit and reject data (opted-in only). Test: a tuned model beats the base on human preference ratings on a blind set.
- [ ] **M19.7** Human evaluation panel and rubric (meaning kept, reads naturally, fits context) with a simple review tool. Test: inter-rater agreement is measured and recorded.
- [ ] **M19.8** Serving stack (vLLM or TGI for decoders, Triton or ONNX for encoders) with batching, quantisation and autoscaling. Test (Load): meets latency targets at the planned peak.
- [ ] **M19.9** Shadow, canary (5 percent) and full rollout with automatic rollback on metric drops. Test: a seeded regression triggers rollback.
- [ ] **M19.10** Drift and freshness monitoring: new machine-text samples from newly released models added on a schedule; detector scored on them. Test: a monthly report lists detector performance on the newest samples.
- [ ] **M19.11** Cost tracking per model and per document. Test: the cost dashboard matches the cloud bill within a small margin.
- [ ] **M19.12** Model cards for every released model (intended use, data, limits, metrics by group). Test: release is blocked without a card.

M20. Test strategy and release gates

Goal: one repeatable way to prove a build is safe to ship. Every module's items link to a test here; a release is allowed only when every gate below is green.

Test layers

Layer	Tooling	Runs when
Unit	pytest and Vitest	Every commit
API contract and integration	OpenAPI-driven tests against a running stack	Every pull request
Fidelity corpus	Custom runner comparing XML and rendered pages	Every pull request that touches the engine, nightly otherwise
Model evaluation	Evaluation harness on frozen sets	Every model change and weekly
E2E	Playwright with accessibility checks	Every pull request and before release
Load	k6 or Locust with realistic documents	Weekly and before release
Security	Scanners, authorisation matrix, annual penetration test	Every pull request, weekly, annually

Release gates

- [ ] **M20.1** All unit, API and E2E tests pass.
- [ ] **M20.2** Fidelity corpus: zero untouched-XML differences; page-count and visual-diff thresholds met.
- [ ] **M20.3** Detector gates: false-positive rate under 1 percent at the top band for every writer group; calibration within limit.
- [ ] **M20.4** Similarity gates: precision and recall at or above target on the test set; a document compared with itself scores near 100 percent; coverage statement present.
- [ ] **M20.5** Rewrite gates: meaning and constraint pass rates at target; zero new-fact failures in the seeded set.
- [ ] **M20.6** Trust gates: precision at each ladder level at target; appeals path works end to end.

- [ ] **M20.7** Performance gates: open in under 4 seconds, first scores in under 5 seconds, report in under 90 seconds for 30 pages, apply under 100 ms.
- [ ] **M20.8** Security gates: no critical or high findings open; authorisation matrix passes.
- [ ] **M20.9** Billing gates: each plan purchased, upgraded, downgraded, failed and refunded in the sandbox; webhooks idempotent.
- [ ] **M20.10** Accessibility gate: WCAG 2.2 AA checks pass on editor, report and dashboards.
- [ ] **M20.11** Staging smoke run with ten real documents, canary at 5 percent, then full rollout with rollback ready.
- [ ] **M20.12** Release notes, migration notes and runbook updates merged.

Launch checklist

Tick these in order in the two weeks before public launch.

Product

- [ ] The core loop works on ten real documents you did not use during development: upload, score, similarity report, fix, export.
- [ ] Reports show the coverage statement and the notice that scores and similarity are estimates, not proof.
- [ ] Pricing page matches the live plans, limits and the pay-per-paper rule.
- [ ] Help centre contains the user manual and the troubleshooting table.

Business

- [ ] Payment processor in live mode; a real $5 test purchase and refund completed.
- [ ] Terms, privacy policy, responsible-use terms and refund policy are published and linked at sign-up.
- [ ] Support inbox, abuse inbox and takedown contact are monitored.
- [ ] Brand, domain, email domain (SPF, DKIM, DMARC) and status page are set up.

Operations

- [ ] Autoscaling tested at twice the expected launch traffic.
- [ ] Cost alerts set at the plan-level cost limit.
- [ ] Backups and a restore drill completed in the last week.
- [ ] On-call rota and incident runbook are in place.
- [ ] Rollback for the app and for each model version rehearsed.

Safety

- [ ] Penetration test fixes verified.
- [ ] Abuse reviewers trained on the playbook and the appeal process.
- [ ] Marketing copy reviewed: no claims of being undetectable or of proving authorship or plagiarism.
- [ ] Closed beta feedback triaged; every severity-one issue closed.